

A COMPARATIVE STUDY OF LIDAR, VISUAL, AND LIDAR-VISUAL FUSION SLAM FOR INDOOR MOBILE ROBOT

1st Lim Jun Yi

*Dept. of Mechatronic Engineering
Faculty of Electrical Engineering & Technology
University Malaysia Perlis (UniMAP)
limjunyi425@gmail.com*

2nd Assoc. Prof. Dr. Shazmin Aniza Abdul Shukor

*Dept. of Mechatronic Engineering
Faculty of Electrical Engineering & Technology
University Malaysia Perlis (UniMAP)
shazmin@unimap.edu.my*

Abstract—Simultaneous Localization and Mapping (SLAM) is a fundamental capability for autonomous indoor mobile robots. This paper presents a comparative evaluation of LiDAR-based SLAM, Visual SLAM, and LiDAR-visual fusion SLAM using identical datasets collected from a TurtleBot-like mobile robot equipped with a 2D LiDAR sensor, an RGB-D camera, and wheel odometry. Experiments were conducted in both real-world and simulation environments within the ROS 2 framework. Localization accuracy was evaluated using Absolute Trajectory Error (ATE), while mapping performance was assessed through qualitative analysis of the generated occupancy maps. The experimental results show that LiDAR-based SLAM consistently achieved the highest localization accuracy and produced the most geometrically consistent maps in the real-world environments. Although LiDAR-visual fusion SLAM improved mapping consistency compared with Visual SLAM, its localization accuracy remained comparable to the visual approach. In the simulation environments, all three SLAM approaches achieved substantially lower localization errors, highlighting the influence of practical sensing conditions on SLAM performance. The findings provide practical guidance for selecting suitable SLAM approaches for consumer-grade indoor mobile robot applications.

Keywords—Simultaneous Localization and Mapping (SLAM), LiDAR-based SLAM, Visual SLAM, LiDAR-visual fusion SLAM, indoor mobile robot, ROS 2

I. INTRODUCTION

Autonomous mobile robots have become increasingly important in applications such as warehouse automation, service robotics, and indoor inspection. To operate reliably in unknown environments, a robot must estimate its position while simultaneously constructing a map of its surroundings, a problem commonly referred to as Simultaneous Localization and Mapping (SLAM). Since Global Navigation Satellite System (GNSS) signals are unreliable in indoor environments [1], SLAM has become a fundamental component for indoor robot navigation.

Numerous SLAM approaches have been developed based on different sensing modalities. Visual SLAM estimates robot motion using image information acquired from monocular, stereo, or RGB-D cameras. It is capable of extracting rich appearance information from the environment, making it suitable for tasks such as loop closure and place recognition. However, its performance is often affected by illumination changes, motion blur, and environments with limited visual features [2]. In contrast, LiDAR-based SLAM estimates robot motion using geometric measurements

obtained from laser range sensors. LiDAR is generally more robust to lighting variations and can produce accurate geometric maps, but it provides limited semantic information and may encounter difficulties in environments containing repetitive structures [3]. To overcome the limitations of individual sensing modalities, LiDAR-visual fusion SLAM combines visual and range information to improve localization robustness and mapping consistency.

Although numerous studies have evaluated different SLAM approaches, direct comparison remains challenging because many experiments are conducted using different robot platforms, sensor configurations, datasets, and evaluation methodologies [4]. This paper presents a comparative evaluation of three representative SLAM approaches for indoor mobile robots: LiDAR-based SLAM, Visual SLAM, and LiDAR-visual fusion SLAM. An indoor mobile robot equipped with a 2D LiDAR sensor, an RGB-D camera, and wheel odometry was used to collect datasets in both real-world and simulated indoor environments. All SLAM algorithms were executed using the same recorded datasets within the ROS 2 framework to ensure consistent experimental conditions. Performance was evaluated based on map quality and localization accuracy using Absolute Trajectory Error (ATE).

II. LITERATURE REVIEW

A. Visual SLAM

Visual SLAM has been widely adopted for indoor mobile robots because cameras provide rich appearance information that facilitates feature extraction, loop closure, and place recognition [5]. Depending on the sensing configuration, Visual SLAM can be implemented using monocular, stereo, or RGB-D cameras. Among these, RGB-D-based approaches are particularly suitable for indoor environments because depth information is directly available, simplifying map construction and scale estimation [6]. However, the performance of Visual SLAM remains highly dependent on environmental conditions. Illumination variation, motion blur, and texture-poor environments may reduce the number of reliable visual features, resulting in localization drift or tracking failure [7].

B. LiDAR-Based SLAM

LiDAR-based SLAM estimates robot motion using geometric range measurements and has become a popular solution for indoor mobile robot localization because of its robustness to lighting conditions and accurate distance measurements. Compared with camera-based approaches, LiDAR sensors provide reliable geometric information even in poorly illuminated environments. Modern graph optimization methods such as Cartographer [8], Karto SLAM [9], and SLAM Toolbox [10] have demonstrated high mapping accuracy and effective loop closure in indoor environments. Nevertheless, LiDAR sensors provide limited appearance information and may experience localization ambiguity in repetitive indoor structures where geometric features are insufficient [11].

C. LiDAR-Visual Fusion SLAM

To exploit the complementary characteristics of cameras and LiDAR sensors, LiDAR-visual fusion SLAM integrates visual appearance information with geometric range measurements. The additional visual constraints improve loop closure and place recognition, while LiDAR measurements provide robust geometric localization under varying illumination conditions [12]. Although fusion approaches generally improve mapping consistency compared with single-sensor methods, they require accurate sensor calibration, time synchronization, and greater computational resources, increasing overall system complexity [13].

D. Research Gap

Although numerous studies have investigated Visual SLAM, LiDAR SLAM, and LiDAR-visual fusion SLAM, direct comparison among these approaches remains challenging because many studies employ different robot platforms, sensor configurations, datasets, and evaluation methodologies. Furthermore, benchmark datasets are commonly collected using high-end synchronized sensors, which may not accurately represent the sensing capabilities of affordable indoor mobile robots. Consequently, the reported performance may not reflect the practical behaviour of different SLAM approaches on consumer-grade robotic platforms. To address this limitation, this study performs a comparative evaluation of representative Visual, LiDAR-based, and LiDAR-visual fusion SLAM algorithms using identical datasets collected from the same TurtleBot-like mobile robot under both real-world and simulation environments.

III. METHODOLOGY

A. Experimental Workflow

Fig. 1 illustrates the overall project flowchart adopted in this study. The methodology consists of five main stages: robot platform setup, experimental environment preparation, data collection, SLAM execution, and performance evaluation. First, a TurtleBot-like mobile robot equipped with a 2D LiDAR sensor, an RGB-D camera, wheel odometry, and an IMU was configured for indoor data acquisition. A corridor and a bedroom were prepared in both real-world and simulation settings. The robot was manually teleoperated

along predefined trajectories while sensor data were synchronously recorded as ROS 2 bag files. The recorded datasets were subsequently processed offline using three representative SLAM approaches: LiDAR-based SLAM, Visual SLAM, and LiDAR-visual fusion SLAM. Since all algorithms were executed using the same recorded datasets, identical sensing conditions were maintained throughout the evaluation. Finally, the estimated trajectories and generated occupancy maps were analysed using Absolute Trajectory Error (ATE) and qualitative map assessment to compare the localization and mapping performance of the evaluated SLAM approaches.

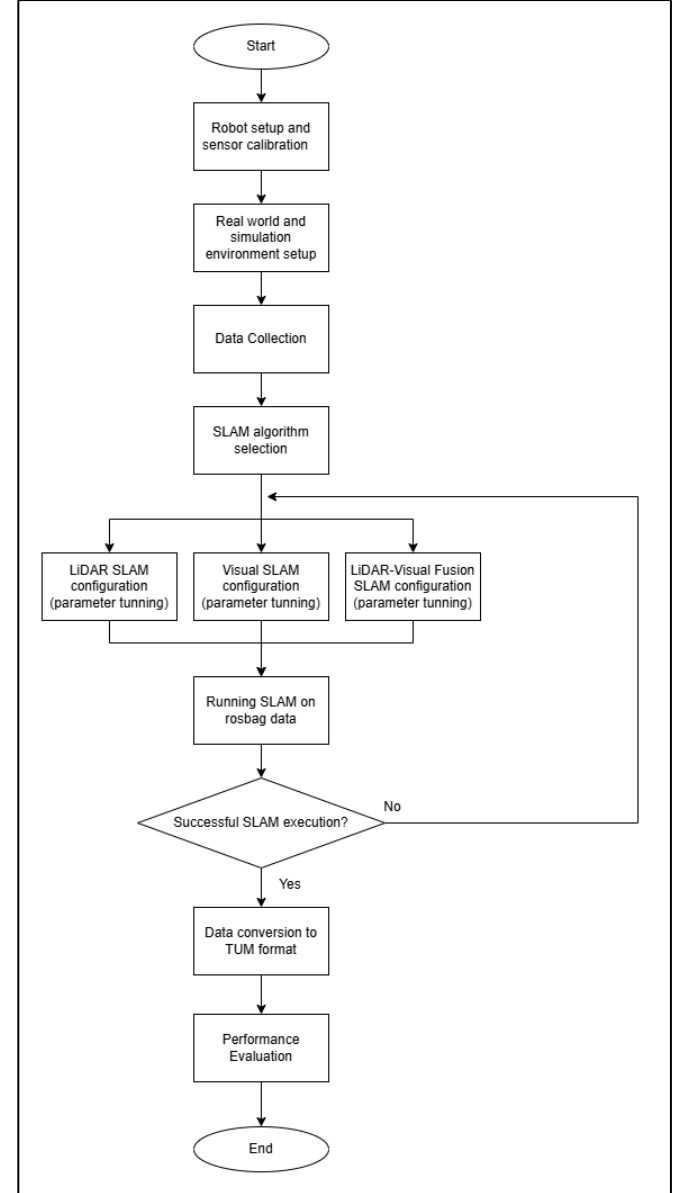


Fig. 1. Project Flowchart

B. Robot Platform

A TurtleBot-like differential-drive mobile robot as shown in Fig. 2. was developed for this study. The robot was equipped with a consumer-grade 2D A2M8 RPLiDAR sensor, an Intel RealSense D415 RGB-D camera, wheel odometry, and an inertial measurement unit (IMU). Sensor data were acquired using a Raspberry Pi 4 running Ubuntu 22.04 LTS and ROS 2 Humble, while an Arduino Uno was

employed for motor control and encoder acquisition. Due to the limited computational capability of the onboard computer, all datasets were recorded as ROS 2 bag files and processed offline on an external workstation. This offline evaluation ensured that all SLAM algorithms were executed using identical sensor data, thereby providing a fair comparison under consistent experimental conditions.

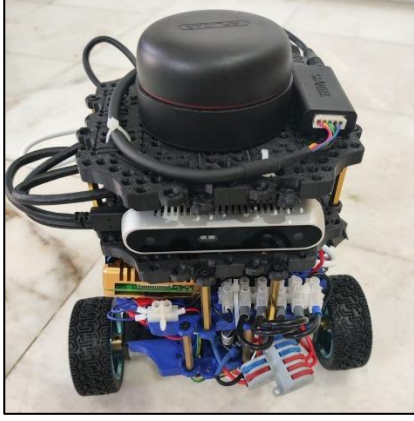


Fig. 2. Mobile Robot

C. Experimental Environment and Data Collection

This study employed both real-world and simulation environments to evaluate the performance of the selected SLAM approaches under consistent experimental conditions. A corridor and a bedroom were selected because they exhibit different structural characteristics commonly encountered in indoor mobile robot applications. The corridor represents a long and relatively feature-sparse environment with repetitive geometric structures, whereas the bedroom contains a higher density of furniture and visual features. To ensure a fair comparison, identical navigation trajectories were performed in both real-world and simulation environments, and all sensor data were recorded as ROS 2 bag files for offline processing by the evaluated SLAM algorithms.

1. Real-World Experimental Environment

Fig. 3 illustrates the layouts of the corridor and bedroom used in the real-world experiments. The corridor provides long straight walls with repetitive structures and limited visual features, making reliable localization and loop closure more challenging. In contrast, the bedroom contains furniture and objects with richer geometric and visual information that can support feature extraction and map construction.

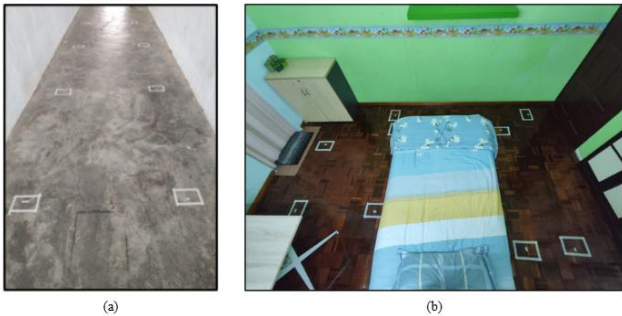


Fig. 3. Real-World Experimental Environment: (a) Corridor, (b) Bedroom

2. Simulation Experimental Environment

Fig. 4 presents the simulation environments developed for this study. They were developed in Gazebo using the measured dimensions and layouts of the physical environments.

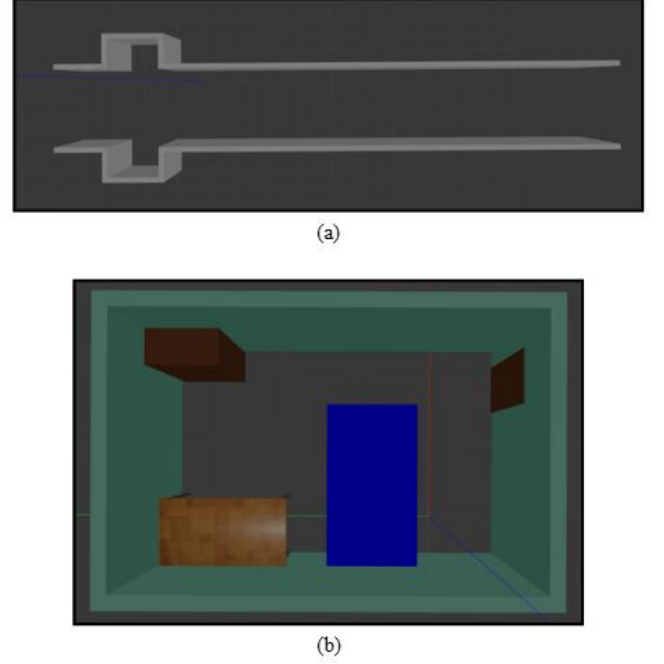


Fig. 4. Simulation Experimental Environment: (a) Corridor, (b) Bedroom

3. Sparse Ground Truth Measurement

Accurate trajectory evaluation requires a reference trajectory for comparison. Since high-precision indoor positioning systems such as motion capture systems were unavailable, a sparse ground truth measurement method was adopted for the real-world experiments. Several reference waypoints were established within each environment as shown in Fig. 5, and their coordinates were measured manually relative to the robot's starting position using a tape measure. During data collection, the robot was navigated to each waypoint and remained stationary for approximately five seconds before continuing along the predefined trajectory. These stationary poses served as reference points for constructing the ground truth trajectory used in the localization accuracy evaluation. For the simulation experiments, the robot's ground truth trajectory was obtained directly from the Gazebo simulation environment.

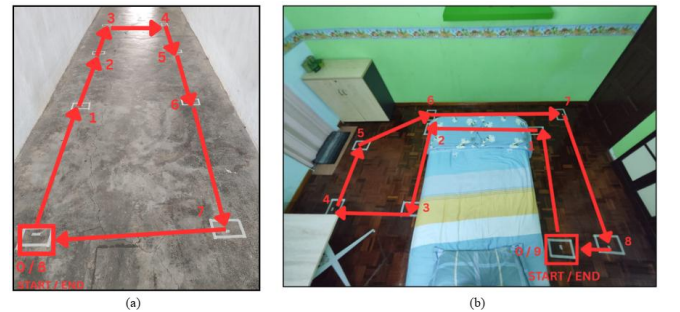


Fig. 5. Experimental Trajectory: (a) Corridor, (b) Bedroom

D. SLAM Algorithm Selection

Three representative SLAM approaches were selected to evaluate the performance of different sensing modalities for indoor mobile robot localization and mapping. LiDAR-based

SLAM was implemented using SLAM Toolbox, which estimates robot poses by matching successive 2D laser scans and performing pose graph optimization. Visual SLAM was implemented using RTAB-Map operating in RGB-D mode, where RGB images and depth information were utilized for visual odometry, loop closure detection, and map construction. For the LiDAR-visual fusion approach, RTAB-Map was implemented to process synchronized RGB-D images and 2D LiDAR scans simultaneously, allowing geometric and visual information to be integrated within a unified mapping framework. Table 1 shows the selected SLAM algorithms.

TABLE I. SELECTED SLAM ALGORITHM

SLAM Category	Algorithm	Sensor Input	Key Strength
LiDAR-based SLAM	SLAM Toolbox [10]	2D LiDAR, Robot Odometry (wheel encoder + imu)	Robust mapping in indoor environments, loop closure, computationally efficient
Visual SLAM	RTAB-Map (RGB-D) [14]	RGB-D Camera, Robot Odometry (wheel encoder + imu)	Rich visual feature extraction, reliable relocalization, direct scale estimation from depth
LiDAR-Visual Fusion SLAM	RTAB-Map (RGB-D + LiDAR fusion) [14]	RGB-D Camera, 2D LiDAR, Robot Odometry (wheel encoder + imu)	Combines geometric accuracy of LiDAR with visual loop closure robustness.

E. SLAM Algorithm Selection

The performance of the evaluated SLAM approaches was assessed using both quantitative and qualitative evaluation methods. Localization accuracy was quantified using Absolute Trajectory Error (ATE), which measures the deviation between the estimated robot trajectory and the corresponding reference trajectory after trajectory alignment. In addition to trajectory accuracy, the quality of the generated occupancy maps was evaluated through qualitative visual inspection. The evaluation focused on wall alignment, structural consistency, loop closure performance, and overall map completeness.

IV. RESULTS & DISCUSSION

A. Real-World Experimental Results

The real-world experiments were conducted using the datasets collected from the corridor and bedroom environments described in Section III. The performance of the evaluated SLAM approaches was assessed through qualitative analysis of the generated occupancy maps and quantitative evaluation of localization accuracy using Absolute Trajectory Error (ATE). Since all SLAM approaches were executed using identical datasets and evaluation procedures, the results provide a direct comparison of their localization and mapping performance under practical indoor operating conditions.

1. Map Quality Evaluation

Table II presents the map quality assessment for the real-world experiments. Overall, LiDAR-based SLAM produced the most geometrically consistent maps, while Visual SLAM exhibited the largest mapping distortion. The LiDAR-visual fusion approach improved map consistency compared with Visual SLAM but did not consistently outperform the LiDAR-based approach.

TABLE II. MAP QUALITY ASSESMENT FOR THE REAL WORLD EXPERIMENTS

Environment	LiDAR-Based SLAM	Visual SLAM	LiDAR-Visual Fusion SLAM
Corridor	Excellent	Poor	Good
Bedroom	Excellent	Moderate	Good

Fig 6. shows the maps generated by the LiDAR-based SLAM, Visual SLAM, and LiDAR-visual fusion SLAM approaches in the corridor environment. The LiDAR-based SLAM accurately preserved the straight wall geometry and corridor dimensions with minimal observable distortion. The generated map remained well aligned throughout the navigation trajectory, indicating effective scan matching and successful loop closure. In contrast, Visual SLAM exhibited noticeable wall deformation and accumulated drift, particularly along the return path, resulting in misaligned corridor boundaries. This degradation can be attributed to the limited visual features and repetitive appearance of the corridor, which reduced the reliability of visual feature tracking and place recognition. Although LiDAR-visual fusion SLAM improved structural consistency compared with Visual SLAM, slight wall distortion remained visible near the loop closure region, indicating that the additional visual information provided only limited improvement under these conditions.

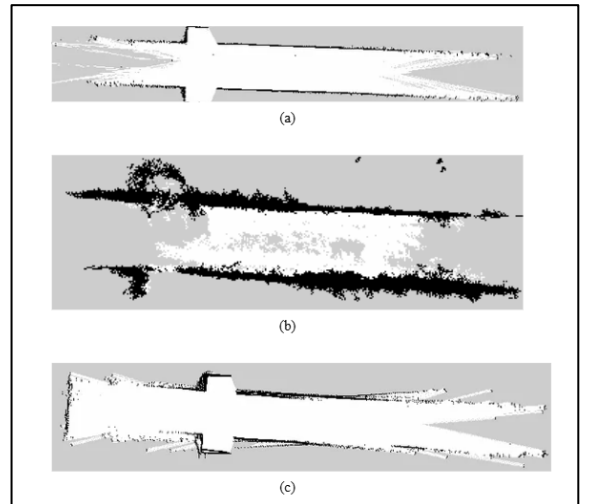


Fig. 6. Real-world Corridor Generated Maps: (a) LiDAR-based SLAM, (b) Visual SLAM, (c) LiDAR-Visual Fusion SLAM

A similar trend was observed in the bedroom environment as shown in Fig. 7. LiDAR-based SLAM generated a complete occupancy map with well-defined walls and furniture boundaries that closely matched the physical environment. The increased number of geometric features within the bedroom further supported reliable scan matching

and reduced localization drift. Visual SLAM benefited from the richer visual texture provided by furniture and indoor objects, producing a more consistent map than in the corridor environment. Nevertheless, minor geometric distortion and wall misalignment were still observed. LiDAR-visual fusion SLAM generated maps that were generally comparable to those produced by LiDAR-based SLAM, although slight inconsistencies remained around object boundaries where visual and LiDAR observations were integrated. Overall, the qualitative evaluation demonstrates that LiDAR-based SLAM consistently achieved the highest mapping quality across both real-world environments.

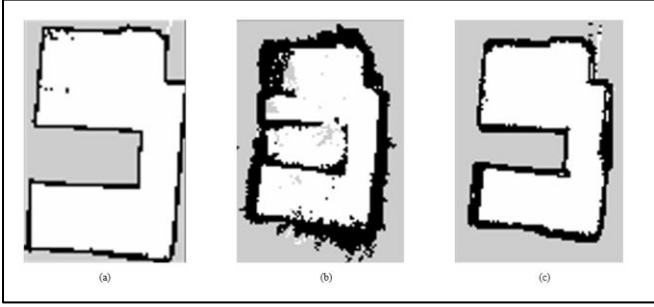


Fig. 6. Real-world bedroom generated maps: (a) LiDAR-based SLAM, (b) Visual SLAM, (c) LiDAR-Visual Fusion SLAM

2. Localization Accuracy Evaluation

Table III shows the ATE results obtained from real world environments. In the corridor environment, LiDAR-based SLAM recorded an ATE RMSE of 6.311 cm, substantially lower than Visual SLAM (18.414 cm) and LiDAR-visual fusion SLAM (17.986 cm). A similar trend was observed in the bedroom environment, where LiDAR-based SLAM achieved an RMSE of 4.250 cm, compared with 13.503 cm for Visual SLAM and 13.501 cm for LiDAR-visual fusion SLAM. These results demonstrate that LiDAR-based SLAM provided the most reliable pose estimation under the evaluated real-world conditions.

TABLE III. REAL-WORLD ATE

Environment	SLAM Approach	ATE RMSE (cm)
Corridor	LiDAR-Based SLAM	6.311
	Visual SLAM	18.414
	LiDAR-Visual Fusion SLAM	17.986
Bedroom	LiDAR-Based SLAM	4.250
	Visual SLAM	13.503
	LiDAR-Visual Fusion SLAM	13.501

The superior localization performance of LiDAR-based SLAM can be attributed to the stability of laser scan matching, which is less sensitive to illumination variation and visual feature availability than camera-based localization. In contrast, the localization accuracy of Visual SLAM was affected by motion blur, image noise, and variations in feature distribution throughout the environments. These factors reduced the robustness of feature matching, leading to increased trajectory drift over longer navigation distances.

Although LiDAR-visual fusion SLAM incorporated both geometric and visual information, its localization accuracy remained comparable to that of Visual SLAM. This suggests that the additional visual constraints provided limited improvement over the LiDAR observations under the evaluated experimental conditions. Possible contributing factors include sensor synchronization uncertainty, calibration errors, and the increased complexity of integrating heterogeneous sensor measurements, which may reduce the effectiveness of the fusion process when using consumer-grade sensing hardware. Fig. 7 shows the real-world environment trajectory comparison

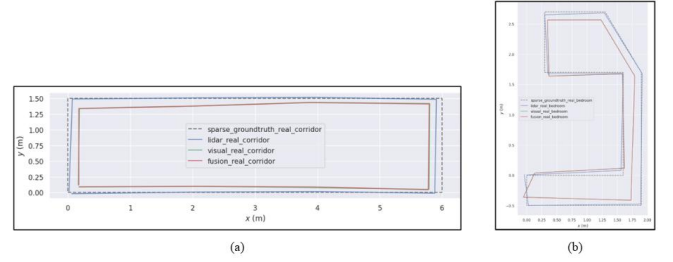


Fig. 7. Real-world Environment Trajectory Comparison: (a) Corridor, (b) Bedroom

B. Simulation Experimental Results

The simulation experiments were conducted using Gazebo environments that replicated the layouts and dimensions of the corresponding real-world corridor and bedroom environments. Since the simulation datasets were generated under controlled conditions without sensor noise, illumination variation, or environmental disturbances, the experiments provide an idealized benchmark for evaluating the intrinsic performance of the selected SLAM approaches.

1. Map Quality Evaluation

Table IV presents the occupancy maps generated by the three SLAM approaches in the simulated corridor and bedroom environments. All evaluated approaches produced maps with high structural consistency and substantially fewer mapping distortions. The controlled simulation environment reduced the influence of measurement noise and sensor uncertainty, allowing each algorithm to operate under more favourable conditions.

TABLE IV. MAP QUALITY ASSESMENT FOR THE SIMULATION EXPERIMENTS

Environment	LiDAR-Based SLAM	Visual SLAM	LiDAR-Visual Fusion SLAM
Corridor	Excellent	Poor	Good
Bedroom	Excellent	Moderate	Excellent

Fig. 8 shows the maps generated by 3 different SLAM approaches in the simulated corridor environment. LiDAR-based SLAM accurately reconstructed the corridor geometry with straight wall boundaries and consistent map alignment throughout the entire trajectory. The Visual SLAM approach generated the least accurate map, exhibiting incomplete wall reconstruction and noticeable mapping artifacts. Next, LiDAR-visual fusion SLAM produced an occupancy map

that closely resembled the LiDAR-based result, with accurate wall alignment and successful loop closure. The qualitative differences among the three approaches were considerably smaller than those observed in the corresponding real-world experiment.

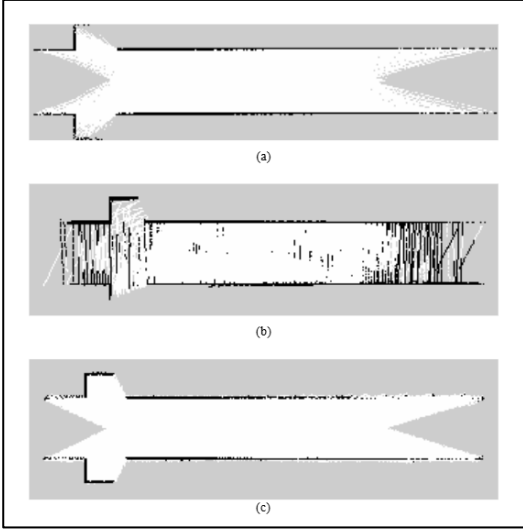


Fig. 8. Simulation Corridor Generated Maps: (a) LiDAR-based SLAM, (b) Visual SLAM, (c) LiDAR-Visual Fusion SLAM

A similar trend was observed in the simulated bedroom environment as shown in Fig. 9. All three SLAM approaches successfully reconstructed the room layout with accurate wall geometry and clearly defined object boundaries. Compared with the real-world experiments, Visual SLAM demonstrated a noticeable improvement in map quality, producing occupancy maps with minimal structural distortion. LiDAR-based SLAM and LiDAR-visual fusion SLAM maintained high mapping consistency throughout the navigation trajectory, resulting in occupancy maps that were visually comparable.



Fig. 9. Simulation Bedroom Generated Maps: (a) LiDAR-based SLAM, (b) Visual SLAM, (c) LiDAR-Visual Fusion SLAM

2. Localization Accuracy Evaluation

Table V shows the ATE results obtained from simulation environments. Compared with the real-world experiments, all evaluated SLAM approaches achieved substantially lower Absolute Trajectory Error (ATE), indicating that localization performance improved considerably under controlled simulation conditions.

TABLE V. SIMULATION ATE

Environment	SLAM Approach	ATE RMSE (cm)
Corridor	LiDAR-Based SLAM	2.085

Bedroom	Visual SLAM	1.354
	LiDAR-Visual Fusion SLAM	1.353
	LiDAR-Based SLAM	1.192
	Visual SLAM	1.224
	LiDAR-Visual Fusion SLAM	1.209
	LiDAR-Based SLAM	1.192

In the simulated corridor environment, Visual SLAM and LiDAR-visual fusion SLAM achieved the lowest localization errors, recording ATE RMSE values of 1.354 cm and 1.353 cm respectively, while LiDAR-based SLAM achieved an RMSE of 2.058 cm. Although the difference among the three approaches was relatively small, these results demonstrate that the availability of ideal RGB-D observations and noise-free feature extraction significantly enhanced the localization performance of the vision-based approaches.

A similar pattern was observed in the simulated bedroom environment, where all three SLAM approaches achieved localization errors of approximately 1 cm. LiDAR-based SLAM recorded an RMSE of 1.192 cm, while Visual SLAM and LiDAR-visual fusion SLAM achieved RMSE values of 1.224 cm and 1.209 cm respectively. The small differences indicate that each approach was capable of accurately estimating the robot trajectory when sensor uncertainty and environmental disturbances were minimized. Overall, the simulation results demonstrate that the performance gap observed among the three SLAM approaches in the real-world experiments was significantly reduced under ideal operating conditions. Fig. 10 shows the simulation environment trajectory comparison.

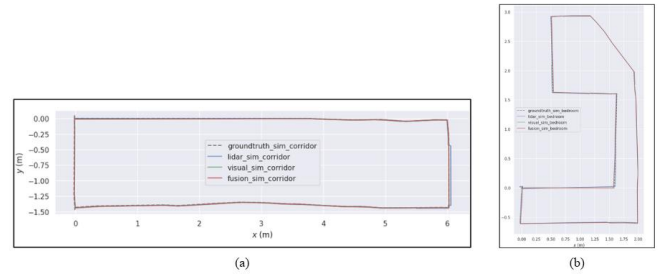


Fig. 10. Simulation Trajectory Comparison: (a) Corridor, (b) Bedroom

C. Comparative Discussion

The experimental results demonstrate that the performance of the evaluated SLAM approaches was strongly influenced by the sensing modality and the operating environment. Although all three approaches successfully generated occupancy maps and estimated robot trajectories, noticeable differences were observed between the real-world and simulation experiments. These findings indicate that environmental conditions and sensor characteristics play a significant role in determining the practical performance of indoor SLAM systems.

LiDAR-based SLAM consistently achieved the best overall performance in the real-world experiments. The generated occupancy maps exhibited high structural consistency, while the corresponding localization errors remained significantly lower than those of the Visual SLAM

and LiDAR-visual fusion SLAM approaches. This superior performance can be attributed to the robustness of laser scan matching, which relies on accurate geometric measurements and is largely unaffected by illumination variation or image quality degradation. Consequently, LiDAR-based SLAM maintained reliable localization even within the feature-sparse corridor environment, where the visual approaches experienced noticeable trajectory drift and mapping distortion.

In contrast, Visual SLAM showed a considerable improvement when evaluated in the simulation environments. Under ideal sensing conditions, the localization accuracy of Visual SLAM became comparable to that of the other evaluated approaches, and the generated occupancy maps exhibited minimal structural distortion. This observation suggests that the relatively poor performance observed in the real-world experiments was primarily caused by practical sensing limitations rather than the visual SLAM algorithm itself. Factors such as image noise, motion blur, rolling shutter effects, illumination variation, and inconsistent feature distribution reduced the reliability of visual feature tracking during the physical experiments, whereas these factors were largely absent in the simulation environment.

Although LiDAR-visual fusion SLAM combined both geometric and visual information, its overall performance did not consistently exceed that of the LiDAR-based approach. While the fusion strategy improved map consistency compared with Visual SLAM, the improvement in localization accuracy remained limited throughout the experiments. A possible explanation is that the LiDAR measurements already provided sufficiently accurate geometric constraints for the relatively structured indoor environments evaluated in this study. Consequently, the additional visual observations contributed only limited complementary information. Furthermore, integrating heterogeneous sensor measurements introduces additional dependencies on accurate calibration, sensor synchronization, and parameter tuning, which may reduce the effectiveness of the fusion process when using consumer-grade sensing hardware.

A comparison between the real-world and simulation results further highlights the influence of sensing conditions on SLAM performance. All evaluated approaches achieved substantially lower localization errors in the simulation experiments, where sensor observations were free from measurement noise, lighting variation, and environmental uncertainty. In contrast, the physical experiments exposed the SLAM algorithms to realistic sensing challenges that more accurately represent practical indoor robot deployment. These findings emphasize that simulation-only evaluations may overestimate the practical performance of SLAM systems, particularly for vision-based approaches that are highly dependent on image quality and environmental texture.

Overall, the results indicate that LiDAR-based SLAM provides the most reliable solution for indoor localization and mapping using consumer-grade sensing platforms under practical operating conditions.

V. CONCLUSION & FUTURE RECOMMENDATION

In conclusion, this paper presented a comparative evaluation of LiDAR-based SLAM, Visual SLAM, and LiDAR-visual fusion SLAM for indoor mobile robot applications using identical datasets collected from a TurtleBot-like mobile robot equipped with a consumer-grade 2D LiDAR sensor, an RGB-D camera, and wheel odometry. To ensure a fair comparison, all SLAM approaches were evaluated under identical sensing conditions in both real-world and simulation environments. Localization performance was assessed using Absolute Trajectory Error (ATE), while mapping performance was evaluated through qualitative analysis of the generated occupancy maps.

The experimental results demonstrated that LiDAR-based SLAM consistently achieved the highest localization accuracy and generated the most geometrically consistent occupancy maps in the real-world experiments. In comparison, Visual SLAM exhibited larger localization errors and mapping distortions, particularly in the corridor environment where limited visual features affected feature tracking and loop closure performance. Although LiDAR-visual fusion SLAM improved mapping consistency relative to Visual SLAM, its localization accuracy remained comparable to the visual approach and did not consistently outperform LiDAR-based SLAM. In the simulation experiments, all evaluated approaches achieved substantially lower localization errors, indicating that the performance differences observed in the real-world experiments were largely influenced by practical sensing challenges such as sensor noise, illumination variation, and feature availability.

Overall, the findings demonstrate that LiDAR-based SLAM provides the most reliable solution for indoor localization and mapping using consumer-grade sensing platforms under practical operating conditions. Furthermore, this study highlights the importance of evaluating SLAM approaches using identical datasets and consistent experimental procedures to ensure a fair and objective comparison. The presented evaluation framework provides useful guidance for selecting suitable SLAM approaches for cost-effective indoor mobile robot applications.

Future work may extend this study by evaluating the selected SLAM approaches in larger and more diverse indoor environments, including dynamic environments containing moving people and obstacles. In addition, incorporating higher-precision ground truth systems, such as motion capture or laser tracking systems, would enable more comprehensive localization accuracy evaluation.

ACKNOWLEDGEMENT

The author would like to extend sincere gratitude to everyone who contributed to the successful completion of this project. Special thanks to the Faculty of Electrical Engineering and Technology at UniMAP for providing the necessary resources and facilities to conduct the study.

REFERENCES

- [1] B. Suleymanoglu, M. Soycan, and C. Toth, "INDOOR MAPPING: EXPERIENCES WITH LIDAR SLAM," in *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives*, 2022. doi: 10.5194/isprs-archives-XLIII-B1-2022-279-2022.
- [2] A. R. Sahili *et al.*, "A Survey of Visual SLAM Methods," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3341489.
- [3] X. Xu *et al.*, "A Review of Multi-Sensor Fusion SLAM Systems Based on 3D LIDAR," 2022. doi: 10.3390/rs14122835.
- [4] M. Filipenko and I. Afanasyev, "Comparison of Various SLAM Systems for Mobile Robot in an Indoor Environment," in *9th International Conference on Intelligent Systems 2018: Theory, Research and Innovation in Applications, IS 2018 - Proceedings*, 2018. doi: 10.1109/IS.2018.8710464.
- [5] G. Jia *et al.*, "Visual-SLAM Classical Framework and Key Techniques: A Review," 2022. doi: 10.3390/s22124582.
- [6] L. Somlyai and Z. Vámosy, "Improved RGB-D Camera-based SLAM System for Mobil Robots," *Acta Polytechnica Hungarica*, vol. 21, no. 8, 2024, doi: 10.12700/APH.21.8.2024.8.6.
- [7] X. Lin *et al.*, "A Robust Keyframe-Based Visual SLAM for RGB-D Cameras in Challenging Scenarios," *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3312062.
- [8] A. Dwijotomo, M. A. A. Rahman, M. H. M. Ariff, H. Zamzuri, and W. M. H. W. Azree, "Cartographer SLAM method for optimization with an adaptive multi-distance scan scheduler," *Applied Sciences (Switzerland)*, vol. 10, no. 1, 2020, doi: 10.3390/app10010347.
- [9] N. T. S. Kumar, M. Gawande, M. N. P. Babu, H. Verma, and P. Rajalakshmi, "Mobile Robot Terrain Mapping for Path Planning using Karto Slam and Gmapping Technique," in *2022 IEEE Global Conference on Computing, Power and Communication Technologies, GlobConPT 2022*, 2022. doi: 10.1109/GlobConPT57482.2022.9938337.
- [10] S. Macenski and I. Jambrecic, "SLAM Toolbox: SLAM for the dynamic world," *J Open Source Softw*, vol. 6, no. 61, 2021, doi: 10.21105/joss.02783.
- [11] T. Guan, Y. Shen, Y. Wang, P. Zhang, R. Wang, and F. Yan, "Advancing Forest Plot Surveys: A Comparative Study of Visual vs. LiDAR SLAM Technologies," *Forests*, vol. 15, no. 12, 2024, doi: 10.3390/f15122083.
- [12] D. Wisth, M. Camurri, S. Das, and M. Fallon, "Unified multi-modal landmark tracking for tightly coupled lidar-visual-inertial odometry," *IEEE Robot Autom Lett*, vol. 6, no. 2, 2021, doi: 10.1109/LRA.2021.3056380.
- [13] D. Van Nam, P. T. Danh, C. H. Park, and G. W. Kim, "Fusion consistency for industrial robot navigation: An integrated SLAM framework with multiple 2D LiDAR-visual-inertial sensors," *Computers and Electrical Engineering*, vol. 120, 2024, doi: 10.1016/j.compeleceng.2024.109607.
- [14] M. Labbé and F. Michaud, "RTAB-Map as an open-source lidar and visual simultaneous localization and mapping library for large-scale and long-term online operation," *J Field Robot*, vol. 36, no. 2, 2019, doi: 10.1002/rob.21831.